Spoken Tutorial – Advanced Prompting Techniques

Title: Advanced Prompting Techniques

Module	Artificial Intelligence Essentials ■ Prompting Skills
Episode	2.2 Advanced Prompting Techniques
Learning Objective	At the end of this tutorial learner will be able to 1. Apply advanced prompting techniques like Chain-of-Thought, structured output, and iterative refinement to enhance AI responses.
Approx. Duration	3-4 min
Outline	• Introduction to Advanced Prompting Techniques. • Using Chain-of-Thought (CoT) for complex problems. • Structuring AI output with specific formats. • Employing Iterative Prompting for refinement. • Summary and Assignment.
Meta Tags	AI prompting, advanced AI, Chain-of-Thought, CoT, structured output, iterative prompting, prompt engineering, ChatGPT, AI skills
Pre-requisite Tutorial	Introduction to Prompting

Script

Visual Cue	Narration
Title slide with 'Advanced Prompting Techniques' as the main title, featuring abstract AI-themed graphics.	Welcome to this Spoken Tutorial. It covers Advanced Prompting Techniques.
Learning Objectives slide with bullet points for Chain-of-Thought, Structured Output, and Iterative Prompting.	In this tutorial, you will learn to apply advanced prompting techniques. These include Chain-of-Thought. You will also learn about structured output. Finally, you will explore iterative refinement.
System Requirements slide showing a generic web browser window on a laptop.	Here, I am using a browser on a computer or mobile.
Prerequisites slide with text 'Familiarity with basic AI prompting'.	To follow this tutorial, you must be familiar with the basics of prompting.
Person at laptop, looking thoughtful, with complex AI response on screen. Represents going beyond basic prompts.	We have learned about clear and concise prompts.
Graphic: 'Chain-of-Thought' title with gears and thought bubbles, 'Think Step-by-Step' highlighted.	First, let's explore Chain-of-Thought, or CoT.
ChatGPT interface, user typing the math problem prompt.	Let's try a math problem without CoT.

ChatGPT response to the math problem, showing a single numerical answer or a slightly incorrect one.	The AI tool might give a direct answer. However, it might be incomplete or incorrect.
Highlighting the phrase 'Think step-by-step' being added to a prompt.	Now, let's add a special phrase. This phrase helps the AI think.
ChatGPT interface, user typing the improved math problem prompt with 'Think step-by-step'.	Type this improved prompt into the AI tool. Then, press enter to see the result.
ChatGPT response showing the step-by-step reasoning for the apple problem.	Observe the AI's response now. Notice the detailed steps.
Graphic: Icons for different output formats (table, list, code, JSON) with a blueprint icon, symbolizing organization.	Next, let's learn to structure the AI's output.
ChatGPT interface, user typing the prompt for a Markdown table.	Let's ask for pros and cons of remote work.
ChatGPT output showing the pros and cons of remote work in a Markdown table.	See how the AI provides a clean table.
Examples of different output formats (numbered list, HTML snippet, JSON object, comma-separated list) on screen.	You can ask for other formats too. For example, try a numbered list or JSON.
Two people conversing, with speech bubbles showing back-and-forth refinement, symbolizing iterative prompting.	Finally, let's talk about Iterative Prompting.
ChatGPT interface, user typing the first slogan prompt and AI responding.	Let's try to get a slogan for a coffee shop.
ChatGPT interface, user typing the follow-up prompt and AI responding with the refined slogan.	That's a good start. However, we can improve it further.
User smiling at the screen, showing the final refined slogan 'Urban Brews. Your Daily Grind.'.	This response is perfect! It meets all our requirements.
Summary slide with icons representing CoT, structured output, and iterative prompting.	With this, we come to the end of this tutorial. Let us summarise the key points covered. We learned about Chain-of-Thought for complex reasoning. We explored structuring AI output into specific formats. And we understood the power of iterative prompting to refine results.
Assignment slide, person working on a laptop, with text overlay of assignment instructions.	As an assignment, please do the following. Find a complex problem online to solve. Try solving it with an AI tool of your choice. First, use a simple prompt for the problem. Then, add "Think step-by-step" and compare the results. Also, try asking the AI to summarize an article as a JSON object.
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